How Much History Is Enough?
Certifying Decision-Relevant Memory in Offline Reinforcement Learning

Anonymous submission

Abstract

Partial observability is often handled by conditioning a policy on a fixed window of recent observations and actions. Longer windows can reveal hidden state, but they enlarge the effective state space and may retain history that improves prediction without changing good decisions. We study how to select the shortest decision-sufficient history window from offline data. Candidate policies use suffixes of the full observable history; importantly, these suffixes need not be Markov. We introduce a cellwise *decision ambiguity*: the minimum worst-case optimal-advantage loss incurred by applying one action distribution to full histories sharing a suffix. Zero ambiguity exactly characterizes the existence of an optimal suffix policy, while cumulative ambiguity sandwiches the best uniform value loss. We develop Decision-Relevant Memory Selection (DRMS), which converts simultaneous confidence intervals for the full-history optimal Q-function into post-selection-valid memory certificates. We further derive a *deployment-weighted* certificate that discounts rare conflicts under an explicit occupancy-ratio condition and remains valid with empirical occupancy estimation. In a canonical rare-context family, exact memory identification requires and, up to logarithms, is achieved with $\Theta(1/(p\gamma^2))$ episodes, where p is the probability of a conflicting history and γ its action gap. Across 4,840 finite-sample datasets and 700 configurations from random history MDPs, the certificates remain calibrated and exhibit the predicted coverage–gap scaling using only lightweight CPU computation.

Introduction

An agent facing partial observability must decide what part of the past to retain. A common practical answer is frame stacking: concatenate a fixed number of recent observations and actions, then apply an ordinary reinforcement-learning algorithm. Choosing this window is consequential. Too little history aliases situations requiring different actions; too much history inflates the effective state space, worsens offline coverage, and invites spurious dependence on nuisance variables. This difficulty is not merely computational: conditioning offline RL directly on long observation histories can have poor sample complexity unless the representation discards information irrelevant to action selection (Hong, Dragan, and Levine 2024). Recurrent policies replace a fixed window with a learned state, but introduce a larger architecture and tuning problem and do not explain how much historical information the task actually requires.

A tempting solution is to select memory through prediction: retain history as long as it improves prediction of the next observation or reward. This tests whether a truncated process is Markov, but Markov sufficiency is stronger than what control requires. Two failures are possible. First, an old nuisance variable may remain highly predictive of future observations even though it never changes the optimal action. Prediction then over-prescribes memory. Second, an old cue may determine which action receives reward while having no effect on subsequent observations. A next-observation test then under-prescribes memory. Figure 1 gives exact instances of both failures.

This paper asks a more direct question:

> *What is the shortest history window for which offline data can certify a near-optimal policy?*

We treat complete observable history as a finite-horizon MDP and compare nested policy classes that use only an m-step suffix. We never assume that the suffix itself is Markov. Instead, we ask whether full histories merged by the suffix can share an action without losing value. This yields a population quantity defined from full-history optimal advantages. It is zero exactly when a uniformly optimal m-memory policy exists and gives an explicit approximation-error certificate when exact sufficiency fails.

The population characterization becomes an offline procedure by replacing unknown advantages with simultaneous upper confidence bounds. Each suffix cell requires only a small minimax linear program. The resulting method, DRMS, either returns a short policy with a high-confidence value-loss bound or abstains when the data do not support one. Because the guarantee holds simultaneously across candidate memories, the same dataset may be used for selection without an additional post-selection penalty.

Uniform protection against every feasible history can nevertheless be too conservative when a conflict is extremely rare. We therefore also derive a deployment-weighted version. It minimizes advantage under a reference occupancy and converts that local quantity into a deployment-value guarantee through an explicit density-ratio bound. An empirical

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correction preserves the guarantee when the reference occupancy is estimated from the same offline data. Thus the paper provides two complementary modes: a distribution-free uniform certificate and a less conservative deployment-specific certificate.

Our contributions are:

- We define *decision ambiguity* for temporal suffix classes and prove monotonicity, an exact decision-sufficiency characterization, and a two-sided value-loss sandwich.
- We prove that next-observation predictive memory and decision-relevant memory are incomparable, and give a closed-form example that exposes the action compromise measured by our criterion.
- We introduce DRMS, a simultaneous interval-based offline certifier that remains valid even when every candidate short suffix is non-Markov. We give an adaptive localization guarantee for approximate memory budgets.
- We extend the framework to occupancy-weighted ambiguity, including a robust empirical-occupancy correction and a concentrability-based deployment guarantee.
- We establish matching dependence $1/(p\gamma^2)$ for exact memory identification in a rare-conflict family.
- We validate the theorem mechanisms on 4,840 finite-sample datasets and 200 random history MDPs. The empirical 50%-recovery threshold has log–log slope 0.935 against $1/(p\gamma^2)$, and no theorem violation is observed in 700 randomized memory configurations.

Related Work

Finite memory under partial observability. General POMDP planning uses a belief over latent states (Kaelbling, Littman, and Cassandra 1998). Finite-state controllers offer compact policy memory (Amato, Bonet, and Zilberstein 2010); known-model analyses quantify finite-window approximation under filtering stability (Kara and Yüksel 2022). Provable short-memory RL assumes that a recent window approximately decodes latent state and obtains complexity depending on the window length rather than the full horizon (Efroni et al. 2022). Recent metric belief approximation gives policy-conditional finite-memory loss bounds (Kim 2026). These results explain when a prescribed memory representation can approximate a belief policy, but they do not select the shortest decision-sufficient window from a fixed offline dataset.

Offline POMDP learning has been analyzed through finite-memory Bellman operators and structural assumptions such as undercompleteness (Guo et al. 2022). Safe policy improvement with finite-state controllers assumes a fixed memory architecture and constrains updates using data support (Simão, Suilen, and Jansen 2023). Memory traces replace raw windows by exponentially weighted summaries and provide offline on-policy evaluation bounds (Eberhard, Muehlebach, and Vernade 2025). Recurrent model-free methods can also be strong empirical baselines (Ni, Eysenbach, and Salakhutdinov 2022). Most directly, Hong, Dragan, and Levine (2024) show that raw observation histories can make offline RL sample-inefficient and motivate compact representations preserving features relevant to action selection. We study the complementary model-selection question: within nested temporal suffix classes, which is the shortest class whose decision loss can be certified from a fixed offline dataset? The selected suffix need not be Markov, and the output includes an explicit value-loss certificate rather than only a learned representation.

Predictive representations and state abstraction. Predictive-state representations summarize history through forecasts of future events (Littman, Sutton, and Singh 2001); causal-state representations learn predictive partitions of action–observation histories (Zhang et al. 2019). Self-predictive representation theory unifies many state and history objectives through latent transition prediction (Ni et al. 2024). A recent Markov Violation Score diagnoses non-Markov observations by testing whether history improves prediction (Mysore 2026). Temporal Range instead measures how strongly a trained policy uses different lags through Jacobian sensitivities and compares the result with window ablations (Lafuente-Mercado, Rus, and Rusch 2025). These are natural diagnostic and representation-learning tools, but they do not provide an offline high-confidence value certificate for a data-selected memory. Our counterexamples further show that observation prediction alone is neither necessary nor sufficient for choosing policy memory.

The state-abstraction literature instead asks which state distinctions preserve value or policy (Li, Walsh, and Littman 2006; Abel, Hershkowitz, and Littman 2016; Abel et al. 2020), and action-sufficient representation learning targets variables needed for control (Huang et al. 2022). History-dependent feature abstractions have also been analyzed as controllers for fully observed MDPs (Patil, Mahajan, and Precup 2024). We specialize this decision-preservation idea to temporal suffix partitions and add a statistical problem absent from the population abstraction view: choosing the suffix from offline data with a simultaneous finite-sample certificate and a coverage-dependent lower bound.

Offline RL, pessimism, and model selection. Offline RL uses pessimism to avoid unsupported decisions (Jin, Yang, and Wang 2021; Shi et al. 2022) and connects naturally to imitation under limited coverage (Rashidinejad et al. 2021). Model-based selection can exploit policy-local model accuracy (Dong et al. 2023), while ModBE gives oracle inequalities for nested value-function classes (Lee et al. 2022). Our nested objects are policy information sets rather than Bellman model classes: a short suffix may be non-Markov and Bellman incomplete yet still contain an optimal policy. We estimate only one full-history confidence set and use it to certify the action compromises created by every suffix partition.

Setting and Objective

Consider a finite-horizon controlled process with stages $h \in [H]$, bounded rewards $R_h \in [0, 1]$, and finite action sets $\mathcal{A}_h$. Let $X_h \in \mathcal{X}_h$ be the complete observable action–observation history before action A_h. Complete history is a valid Markov

Approach	Data regime	Memory object	Primary target	Guarantee relevant here
Short-term memory (Efroni et al. 2022)	Online	Fixed observation window	Learn under latent-state decodability	Regret/sample complexity for a chosen window
Offline POMDP RL (Guo et al. 2022)	Offline	Fixed finite memory	Value learning under structural assumptions	Policy suboptimality with prescribed memory
Safe FSC improvement (Simão, Suilen, and Jansen 2023)	Offline	Fixed controller memory	Improve a behavior controller safely	High-probability improvement under support constraints
Memory traces (Eberhard, Muehlebach, and Vernade 2025)	Offline evaluation	Fixed trace parameters	Compress history for value prediction	Evaluation error for trace representations
History representations (Hong, Dragan, and Levine 2024)	Offline	Learned compact history feature	Improve sample complexity of history-based RL	Sufficient representation conditions; bisimulation objective
Temporal diagnostics (Mysore 2026; Lafuente-Mercado, Rus, and Rusch 2025)	Logged trajectories / trained policy	History score	Detect non-Markovity or quantify lag use	Predictive or sensitivity diagnostic; window guidance
DRMS (ours)	Offline	**Selected suffix length**	**Preserve decisions within a loss budget**	**Post-selection uniform or deployment-weighted value certificate**

Table 1: The closest lines of work differ in whether memory is fixed or selected and whether sufficiency is defined by prediction, latent-state recovery, or decision loss.

state, with reward mean $r_h(x,a)$ and transition kernel $P_h(\cdot \mid x,a)$. Let Q_h^*, V_h^* be the corresponding optimal action-value and value functions.

For memory length m, the map $\phi_{m,h}(x)$ retains the latest m observations and intervening actions, or all available history when $h < m$. A policy belongs to

$$\Pi_m = \{\pi : \pi_h(\cdot \mid x) = \pi_h(\cdot \mid \phi_{m,h}(x))\}.$$

The classes are nested, $\Pi_1 \subseteq \cdots \subseteq \Pi_H$. We emphasize that $Z_h = \phi_{m,h}(X_h)$ need not be Markov for $m < H$.

Our distribution-free target is the best uniform loss

$$\mathcal{E}_m = \inf_{\pi \in \Pi_m} \max_{h,x} \big[V_h^*(x) - V_h^\pi(x)\big]. \tag{1}$$

For a tolerated loss ϵ, define the oracle memory $m_\epsilon^* = \min\{m : G_m \le \epsilon\}$ after G_m is introduced below. This target differs from recovering a "true" predictive order: several memory lengths may be non-Markov while already supporting near-optimal decisions.

Decision-Relevant Memory

Define the optimal advantage loss $A_h^*(x,a) = V_h^*(x) - Q_h^*(x,a) \ge 0$. For suffix z, let $C_{m,h}(z) = \{x : \phi_{m,h}(x) = z\}$ be the full histories that an m-memory policy must treat identically.

Definition 1 (Decision ambiguity)**.** *The ambiguity of one suffix cell is*

$$\eta_{m,h}(z) = \min_{p \in \Delta(\mathcal{A}_h)} \max_{x \in C_{m,h}(z)} \mathbb{E}_{a \sim p} A_h^*(x,a). \tag{2}$$

Let $\eta_{m,h} = \max_z \eta_{m,h}(z)$ *and* $G_m = \sum_{h=1}^H \eta_{m,h}$.

The action distribution p is shared by every full history in the suffix cell. Thus $\eta_{m,h}(z)$ measures the smallest unavoidable one-step compromise caused by forgetting distinctions inside that cell. It is a linear program with $|\mathcal{A}_h|+1$ variables. Randomization can reduce worst-case loss, but zero ambiguity has the same meaning for deterministic and stochastic policies: the cell contains a common optimal action.

Example 1 (Two conflicting histories). Suppose a cell contains x_0, x_1 and two actions. Their advantage rows are $(0,\alpha)$ and $(\beta, 0)$. If action zero is chosen with probability q, the two losses are $\alpha(1-q)$ and βq. The minimax solution equalizes them:

$$q^* = \frac{\alpha}{\alpha+\beta}, \qquad \eta = \frac{\alpha\beta}{\alpha+\beta}. \tag{3}$$

For a symmetric conflict $\alpha = \beta = \gamma$, forgetting costs $\gamma/2$. Under occupancy weights $(1-p, p)$, the corresponding expected cost is instead $\min\{(1-p)\alpha, p\beta\}$, foreshadowing the weighted extension.

Theorem 1 (Population characterization)**.** *For every m:*

1. $G_{m+1} \le G_m$.
2. $G_m = 0$ *if and only if there exists* $\pi \in \Pi_m$ *with* $V_h^\pi(x) = V_h^*(x)$ *for every* h, x.
3. *The best attainable uniform loss satisfies*

$$\max_h \eta_{m,h} \le \mathcal{E}_m \le G_m. \tag{4}$$

The cellwise minimax policy π^m *additionally obeys* $V_h^*(x) - V_h^{\pi^m}(x) \le \sum_{t=h}^H \eta_{m,t}$.

Proof idea. Longer memory refines every suffix cell, so the coarser action rule remains feasible and ambiguity cannot increase. For the upper bound, let $D_h(x) = V_h^*(x) - V_h^{\pi^m}(x)$. Adding and subtracting the full-history optimal continuation gives

$$\begin{aligned} D_h(x) &= \mathbb{E}_{a\sim\pi_h^m} A_h^*(x,a) + \mathbb{E}_{a,X'} D_{h+1}(X') \\ &\le \eta_{m,h} + \|D_{h+1}\|_\infty. \end{aligned} \tag{5}$$

Backward induction proves the cumulative bound. Conversely, any suffix policy must use one action distribution throughout each cell; replacing its continuation by V^* can only increase its value, so some context incurs at least the cell minimax loss. Nonnegativity makes $G_m = 0$ equivalent to a common zero-advantage action distribution in every cell. Full proofs are in the supplement.

For behavior policy μ, write $Z_h^m = \phi_{m,h}(X_h)$ and define the oracle next-observation residual

$$\begin{aligned} R_m^{\text{obs}} = &\sum_h H_\mu(O_{h+1} \mid Z_h^m, A_h) \\ &- \sum_h H_\mu(O_{h+1} \mid X_h, A_h). \end{aligned} \tag{6}$$

This mirrors observation-prediction diagnostics, not the stronger condition that reward and controlled suffix transitions are identical within every cell. The latter condition implies $G_m = 0$ by backward induction; the converse need not hold.

Proposition 1 (Prediction–decision separation)**.** *For any horizon H, there are controlled processes under a uniform behavior policy in which: (i) $R_m^{\text{obs}} = 0$ for every m, but $G_m = \gamma/2$ for all $m < H$ and $G_H = 0$; and (ii) $G_m = 0$ for every m, but $R_m^{\text{obs}} = 1$ bit for all $m < H$. Thus an observation-prediction diagnostic can require arbitrarily more or less history than decision sufficiency.*

Offline Certification with DRMS

Assume an offline procedure produces simultaneous intervals

$$L_h(x,a) \le Q_h^*(x,a) \le U_h(x,a) \quad \text{for all } h, x, a \tag{7}$$

with probability at least $1-\delta$. The result is modular: any valid full-history Q^* intervals may be used.

A naive upper advantage, $\max_b U_h(x,b) - L_h(x,a)$, is unnecessarily loose because comparing action a with itself has exactly zero loss. Define

$$\overline{A}_h(x,a) = \max\left\{0, \max_{b\neq a}[U_h(x,b) - L_h(x,a)]\right\}. \tag{8}$$

On event (7), $A_h^*(x,a) \le \overline{A}_h(x,a)$. Replacing A_h^* by $\overline{A}_h$ in Eq. (2) gives robust cell ambiguities $\overline{\eta}_{m,h}(z)$, total certificate $\overline{G}_m$, and robust suffix policy $\overline{\pi}^m$.

Theorem 2 (High-confidence value certificate)**.** *On event* (7)*, simultaneously for every memory m,*

$$V_h^*(x) - V_h^{\overline{\pi}^m}(x) \le \sum_{t=h}^{H} \overline{\eta}_{m,t} \quad \textit{for all } h, x. \tag{9}$$

Consequently, whenever DRMS *certifies $\overline{G}_{\widehat{m}} \le \epsilon$, its data-selected policy has uniform value loss at most ϵ.*

Algorithm 1 Decision-Relevant Memory Selection (DRMS)

Require: Offline trajectories, candidate memories $\mathcal{M}$, tolerance ϵ, confidence δ
- Construct simultaneous full-history intervals (L, U)
- Form upper advantages $\overline{A}$ by Eq. (8)
- **for** $m \in \mathcal{M}$ in increasing order **do**
  - **for** each stage h and suffix cell z **do**
    - Solve $\min_p \max_{x\in C_{m,h}(z)} \langle p, \overline{A}_h(x,\cdot)\rangle$
  - **end for**
  - Sum cell maxima over stages to obtain $\overline{G}_m$
  - **if** $\overline{G}_m \le \epsilon$ **then**
    - **return** m, robust suffix policy $\overline{\pi}^m$, certificate $\overline{G}_m$
  - **end if**
- **end for**
- **return** longest candidate with an explicit abstention flag

The proof repeats Eq. (5) with $A_h^* \le \overline{A}_h$. Simultaneous intervals are essential: the guarantee then applies to all candidate memories on one event, including the memory chosen after inspecting their certificates.

Proposition 2 (Interval tightness)**.** *Suppose every interval has one-sided error at most w. Then*

$$G_m \le \overline{G}_m \le G_m + 2Hw. \tag{10}$$

Corollary 1 (Adaptive localization)**.** *Let $m_\tau^* = \min\{m : G_m \le \tau\}$, and include full history among the candidates. For every $\epsilon \ge 2Hw$,*

$$m_\epsilon^* \le \widehat{m}_\epsilon \le m_{\epsilon-2Hw}^*. \tag{11}$$

If $m_{\text{dec}} = \min\{m : G_m = 0\}$ and $\Gamma = \min_{m<m_{\text{dec}}} G_m > 2Hw$, every $\epsilon \in [2Hw, \Gamma)$ recovers m_{dec}.

Equation (11) is an oracle-style statement for memory parsimony. The selected window is never shorter than the true ϵ-sufficient window, while its excess length is controlled by the common interval resolution $2Hw$. This is not ordinary validation over separately trained models: statistical uncertainty is inherited from one full-history confidence set.

Deployment-Weighted Memory

Uniform ambiguity protects every feasible full history, including events that are negligible under deployment. Let ν_h be a reference distribution over full histories and define

$$\ell_{m,h}^\nu(z,p) = \sum_{x\in C_{m,h}(z)} \nu_h(x)\mathbb{E}_{a\sim p} A_h^*(x,a),$$

$$g_{m,h}^\nu = \sum_z \min_{p\in\Delta(\mathcal{A}_h)} \ell_{m,h}^\nu(z,p), \qquad G_m^\nu = \sum_h g_{m,h}^\nu. \tag{12}$$

Unlike the minimax cell problem, each weighted cell objective is linear, so a deterministic action is optimal except at ties. Partition refinement again gives $G_{m+1}^\nu \le G_m^\nu$.

Theorem 3 (Deployment-weighted certificate)**.** *Let $\pi^{m,\nu}$ minimize Eq.* (12)*, and let $d_h^{\pi^{m,\nu}}$ denote its deployment occupancy from initial distribution ρ. If*

$$d_h^{\pi^{m,\nu}}(x) \le C_h \nu_h(x) \quad \textit{for every } h, x, \tag{13}$$

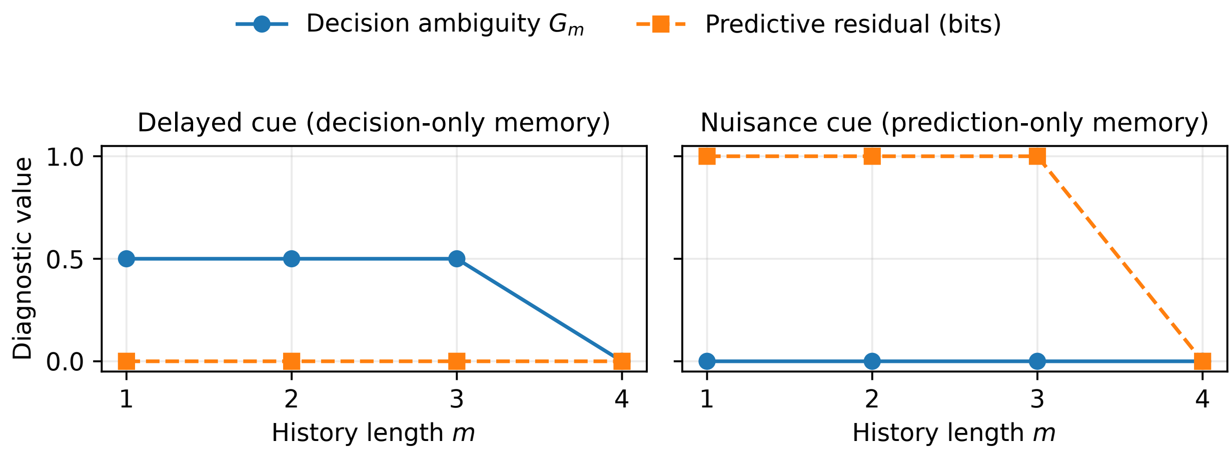

Figure 1: **Prediction and decision memory disagree in both directions.** In the delayed-cue task (left), a stage-one cue determines the final rewarding action, but later observations are constant: $R_m^{\text{obs}} = 0$, whereas $G_m = 1/2$ until the cue enters the suffix. In the nuisance task (right), the cue predicts the terminal observation but never changes the optimal action, so $G_m = 0$ at every memory length. Values are exact for $H = 4$ and a balanced binary cue.

then

$$V_1^*(\rho) - V_1^{\pi^{m,\nu}}(\rho) \le \sum_{h=1}^{H} C_h g_{m,h}^{\nu}. \tag{14}$$

The performance-difference identity writes deployment loss as the sum of optimal advantages under d_h^{π}. Because these costs are nonnegative, Eq. (13) changes measure to ν_h and yields Eq. (14). The assumption is policy-specific: it must hold for the policy returned by the weighted optimization, not merely for a population comparator.

The same idea remains valid offline. Replace A^* by $\overline{A}$, estimate ν_h by $\widehat{\nu}_h$, and suppose $\|\widehat{\nu}_h - \nu_h\|_1 \le \xi_h$ and $0 \le \overline{A}_h \le B_h$. For the robust policy selected using $\widehat{\nu}$, total-variation duality gives the simultaneous correction

$$V_1^*(\rho) - V_1^{\widehat{\pi}_m}(\rho) \le \sum_h C_h \left(\widehat{\overline{g}}_{m,h} + \frac{B_h \xi_h}{2} \right). \tag{15}$$

For n independent trajectories, a Weissman-style occupancy radius is

$$\xi_h = \min \left\{ 2, \sqrt{\frac{2\{|\mathcal{X}_h| \log 2 + \log(H/\delta_{\text{occ}})\}}{n}} \right\}. \tag{16}$$

If no credible finite C_h is available, one should use the uniform certificate. In the rare-cue experiment below, actions do not affect state occupancy, so $C_h = 1$ exactly.

Tabular Instantiation and Statistical Limit

For transparency, we instantiate Eq. (7) in a finite full-history MDP from n independent behavior-policy episodes. Empirical rewards use union-bounded Hoeffding radii; empirical transitions use an L_1 radius based on Weissman et al. (2003). Optimistic and pessimistic dynamic programming propagate these model intervals using $|(P - \widehat{P})^\top f| \le \frac{1}{2}\|P - \widehat{P}\|_1 \operatorname{span}(f)$. Unobserved pairs receive the full reward radius and transition radius two; therefore the method abstains rather than extrapolating through unsupported histories. The supplement states the complete recursion and coverage proof.

Coverage is not merely an artifact of our algorithm. The following two-point family isolates the intrinsic difficulty of deciding whether an old cue is needed.

Theorem 4 (Memory-order lower bound)**.** *Fix $0 < p < 1$, $0 < \gamma \le 1/2$, and $0 < \delta < 1/4$. There are two horizon-two processes observed under the same uniform behavior policy whose shortest decision-sufficient memories are one and two, respectively. Any estimator identifying the correct memory with error at most δ on both requires*

$$n \ge \frac{\log(1/(4\delta))}{6p\gamma^2}. \tag{17}$$

A cue occurs with probability p. In the first model, action zero is better by γ after either cue. In the second, the optimal action flips only after the rare cue. The one-episode KL divergence is at most $6p\gamma^2$; Bretagnolle–Huber then proves Eq. (17) (Tsybakov 2009). Thus memory selection is hard precisely when decision-conflicting histories are rare or their action gaps are small.

Corollary 2 (Matching upper rate)**.** *Let $p \le 1/2$, and let M be the number of full-history state–action pairs. Tabular* DRMS *with tolerance $\gamma/4$ recovers the correct memory in both models with probability at least $1 - \delta$ whenever*

$$n \ge \frac{32 \log(8M/\delta)}{p\gamma^2}. \tag{18}$$

The proof combines a lower tail for rare state–action counts with reward confidence radii small enough to separate the two actions. The lower bound is therefore tight in p and γ, up to constants and logarithms.

Random-history stress-test statistic	Result
Random MDPs	200
Memory configurations	700
Recorded theorem violations	0
95% upper violation rate after 0 failures	0.427%
Median relative slack of G_m	24.6%
90th-percentile relative slack	65.5%
Weighted certificate validity	100%
Robust weighted certificate validity	100%

Table 2: Stress test over random binary-observation, binary-action history MDPs at horizons three and four. "Violation" aggregates monotonicity, both sides of the population sandwich, robust inflation, and uniform and weighted value certificates.

Experiments

Our experiments are deliberately small: their purpose is to test theorem predictions, not to claim deep-control state of the art. Population values are computed exactly. Finite-sample studies use tabular counts, interval dynamic programming, and exact two-action cell solvers. All grids and raw results are included in the repository and supplement.

Prediction selects the wrong memory in both directions. Figure 1 uses horizon four. In the delayed-cue process, a memoryless final action incurs minimax loss $1/2$, while next observations are deterministic. In the nuisance process, the cue contributes one bit of terminal predictive information but action zero is optimal for both cues. The computed policy loss equals G_m in both examples. Varying the delayed-cue gap verifies the exact phase transition from Eq. (3): memory one is ϵ-sufficient exactly when $\gamma \leq 2\epsilon$.

Coverage–gap scaling and calibration. Figure 2 contains 2,880 independent datasets over 12 (p, γ) combinations. The 50%-recovery threshold scales with $1/(p\gamma^2)$ as predicted by Theorem 4. In a separate 1,600-dataset sweep with four p values and fixed $\gamma = 0.5, \epsilon = 0.1$, simultaneous Q-interval coverage and certificate validity are both 100%, and no run falsely certifies a shorter memory. These are calibration checks, not replacements for the theorem.

Deployment weighting. Figure 3 demonstrates the distinction between uniform and expected deployment loss. At $p = 0.05$ and 0.10, the weighted method reaches 100% short-memory certification by 2,500 episodes; at $p = 0.20$, it reaches 100% by 25,000 episodes. At $p = 0.50$, the population weighted cost is 0.2 and memory one is never certified under budget 0.1. Across all 360 weighted runs, Q-interval coverage, occupancy-radius coverage, and the corrected value certificate are 100%. This experiment uses $C_h = 1$; no unknown density ratio is estimated.

Random history MDPs. To move beyond hand-designed cues, we sample rewards from $\mathrm{Beta}(2, 2)$ and fully supported observation transitions from $\mathrm{Dirichlet}(1, 1)$ while keeping complete action–observation histories distinct. Table 2 reports every candidate memory on 100 instances at each of horizons three and four, using synthetic Q^* interval half-width $w = 0.05$. There are no recorded theorem violations. The exact one-sided binomial bound quantifies the resolution of this finite stress test; it is not a proof. Scatter plots and all component checks appear in the supplement.

Computation and reproducibility. The suite contains 4,840 finite-sample datasets, 700 random-MDP memory configurations, and 16 theorem-driven tests. It uses no GPU. The code includes MDP constructors, uniform and weighted ambiguity solvers, interval dynamic programming, raw and summarized CSVs, figure generation, and a one-command paper build. Two-action minimax cells use an exact $O(n \log n)$ upper-envelope solver, cross-checked against generic linear programming in randomized unit tests.

Limitations and Conclusion

The theory enumerates a finite maximum observable history and therefore inherits exponential growth in raw history length. Its value is conceptual and statistical rather than a claim that tabular full-history enumeration scales to rich observations. Function approximation could replace the full-history interval oracle, but obtaining reliable simultaneous Q^* bounds in that setting is itself a major offline-RL problem. The weighted guarantee requires a known or conservatively bounded deployment-to-reference ratio for the *returned* policy; using an optimistic ratio estimate would invalidate the certificate. Our experiments are controlled theorem tests rather than standard POMDP benchmark comparisons.

Within this scope, the results establish a simple principle: memory should be selected according to the action regret created by forgetting, not according to prediction alone. Decision ambiguity makes this principle exact at the population level. DRMS turns it into an offline certificate that remains valid after selecting among memory lengths and can abstain under inadequate coverage. The weighted extension separates rare from uniformly dangerous conflicts, while the matching coverage–gap rate clarifies when any exact memory selector can succeed.

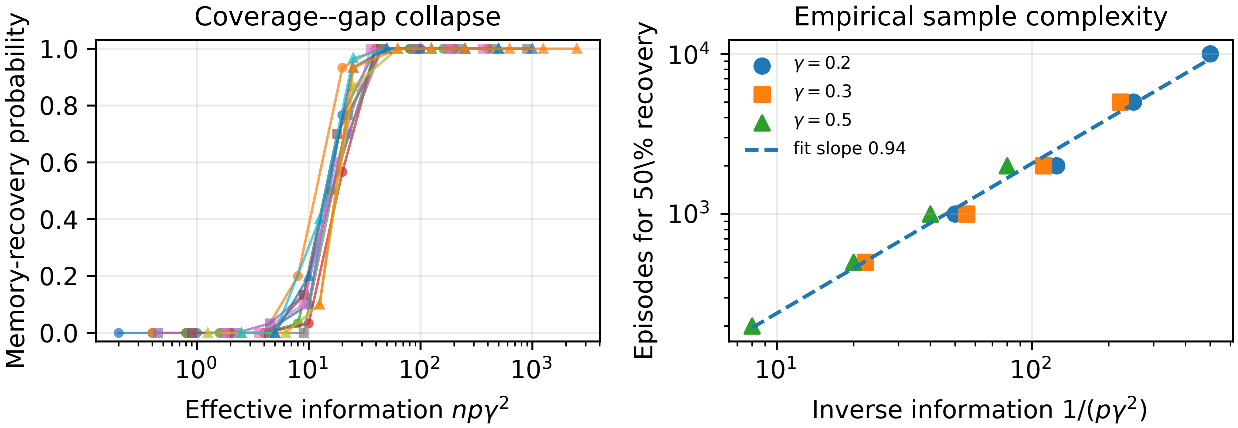

Figure 2: **Finite-sample behavior follows the information-theoretic scaling.** We vary rare-cue probability $p \in \{0.05, 0.1, 0.2, 0.5\}$, action gap $\gamma \in \{0.2, 0.3, 0.5\}$, and eight dataset sizes, with 30 seeds per point. The tolerance is $\gamma/4$. Left: all 12 recovery curves nearly collapse against $np\gamma^2$. Right: the first grid size attaining at least 50% correct certification is approximately linear in $1/(p\gamma^2)$; fitted log–log slope 0.935.

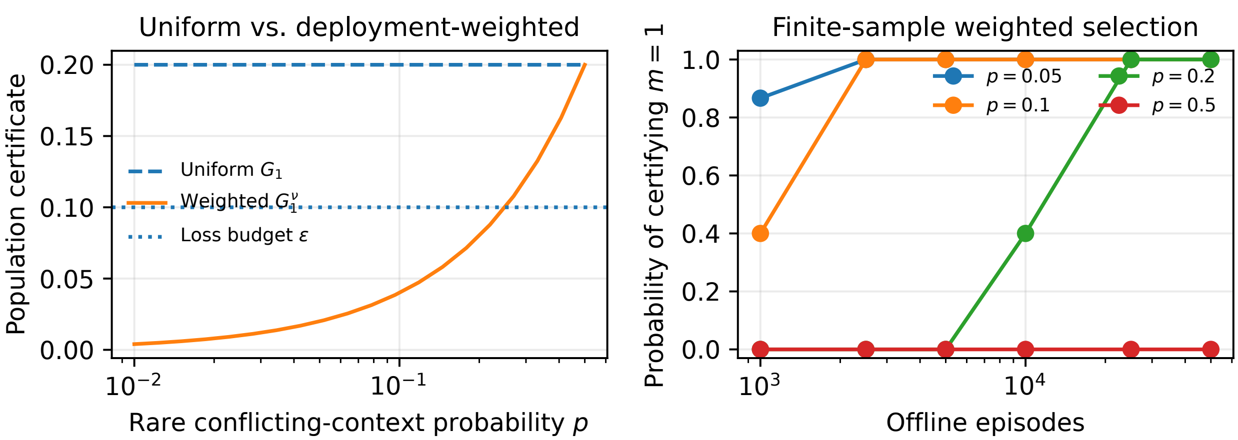

Figure 3: **Deployment weighting avoids unnecessary memory for rare conflicts.** Gap $\gamma = 0.4$ and loss budget $\epsilon = 0.1$. Left: uniform ambiguity is $G_1 = \gamma/2 = 0.2$ for every p, while the deployment-weighted value is $G_1^\nu = p\gamma$ for $p \leq 1/2$. Right: finite-sample robust weighted DRMS certifies memory one with increasing probability for $p \leq 0.2$; it never does so for the balanced conflict $p = 0.5$. Uniform DRMS never certifies memory one in this experiment.

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